

Question	Answer	Marks
2(a)	product of mass and velocity	B1
2(b)	$F = m\Delta v / \Delta t$ $= 16 \times 0.60 / 1.1$	C1
	$= 8.7 \text{ N}$	A1
2(c)	$x = ut + \frac{1}{2}at^2$ $x = 0.60 \times 3.7 + \frac{1}{2} \times 0.85 \times 3.7^2$	C1
	or	
	$v = 0.60 + 0.85 \times 3.7 (= 3.75 \text{ m s}^{-1})$ $x = 3.75 \times 3.7 - 0.5 \times 0.85 \times 3.7^2$ or $x = \frac{1}{2} \times (0.60 + 3.75) \times 3.7$ or $x = (3.75^2 - 0.60^2) / (2 \times 0.85)$	(C1)
	$x = 8.0 \text{ m}$	A1
2(d)(i)	$F = W / s$	C1
	$= 250 / 18$ $= 14 \text{ N}$	A1
2(d)(ii)	any line starting at distance = 0 and a positive non-zero value of kinetic energy	B1
	a straight line from distance = 0 to distance = x with positive gradient	B1
	a straight horizontal line at a non-zero value of kinetic energy starting at distance = x and ending at distance = x + 18 m that is continuous with the previous line	B1

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